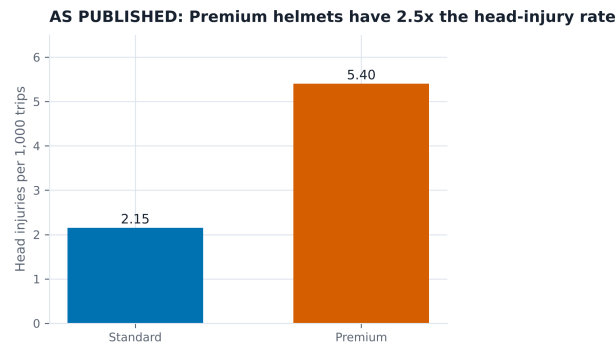


Week 3 - Part A: nine charts that are wrong

Syed Sami Ullah - BSAI-7B · CDB601220 Data Visualization · all data synthetic (CDB601220)

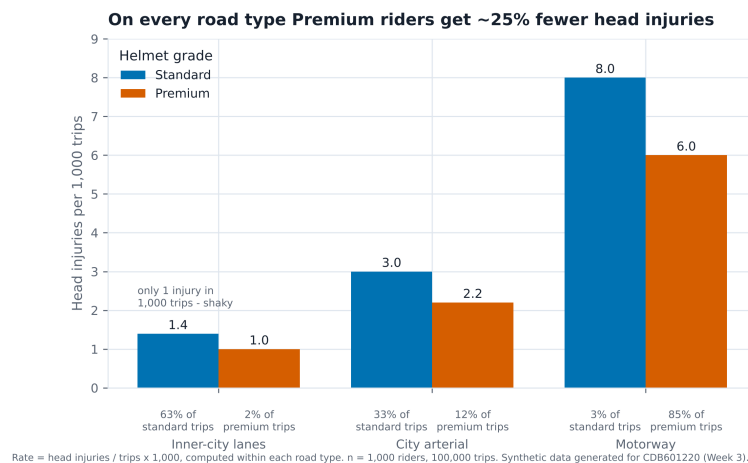
For each case: the rebuilt chart (top, exactly as the recipe says, self-check passed), the honest redesign (below), then the conclusion it invites, the decoy and why it does not hold, the mechanism with the number that proves it, and what the redesign changes. Code: [partA_charts.py](#) / [partA_nine_charts.ipynb](#). All 52 self-checks pass.

Case 1 — “Premium-certified helmets have 2.5x the head-injury rate”



Rebuild of the published chart. Synthetic data generated for CDB601220 (Week 3).

This is the chart the way the analyst drew it. Just looking at it, Premium helmets look 2.5 times worse, because their bar is way taller. But it mixes all three road types together, and the two helmet groups don't ride on the same roads.



Once I split the trips by road type, the story flips. Premium has fewer head injuries on every road, about 25% fewer. It only looked worse before because 85% of Premium trips are on motorways, which is the most dangerous road.

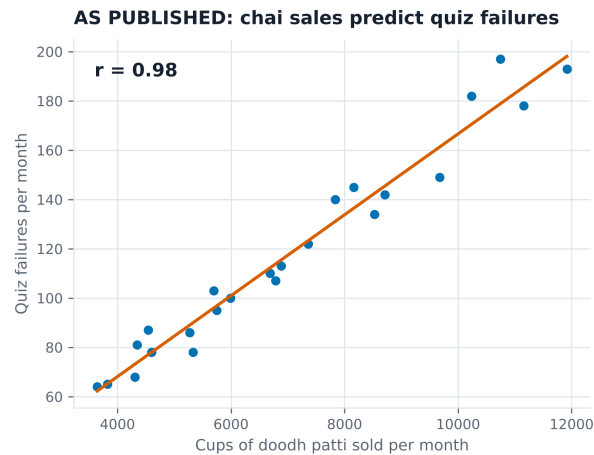
The conclusion it invites. The chart invites me to conclude that Premium helmets are more dangerous than Standard ones: 5.40 head injuries per 1,000 trips against 2.15.

The decoy, and why it does not hold. My first thought was risk compensation: riders who buy the expensive helmet feel safe and ride badly, or they are newer riders. The data rules that out. Average experience is the same in both groups (31.4 vs 31.5 months). Premium riders are faster on average (67 vs 30 km/h), but inside each road type the speeds are almost identical (motorway 72.2 vs 71.0, city arterial 39.6 vs 38.0, inner-city lanes 25.6 vs 24.0). The speed gap is just the road gap.

The mechanism. Simpson's paradox, with road type as the hidden variable. 85% of Premium trips are on motorways, but only 3.3% of Standard trips are. Motorways are by far the most dangerous road (6 to 8 injuries per 1,000 trips, against 1 to 1.4 in the lanes). When I compare the helmets on the same road, Premium is lower on every single one: motorway 6.0 vs 8.0, city arterial 2.2 vs 3.0, inner-city lanes 1.0 vs 1.4. That is roughly 25% fewer injuries with Premium. The headline reverses because the two groups ride on different roads, not because the helmet is worse.

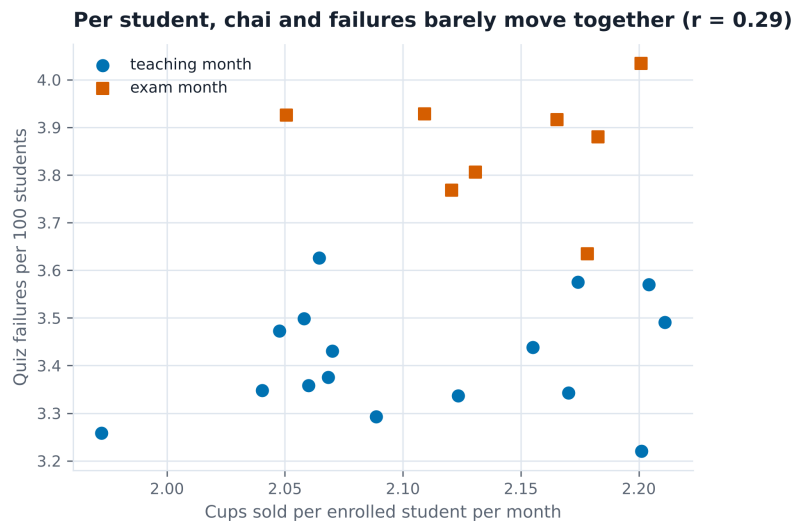
The redesign. One panel per road type, both helmets side by side, with the share of trips written under each bar. The comparison is now like for like, and the reader can see why the pooled number was misleading. (Premium in the lanes is only 1 injury in 1,000 trips, so that bar is shaky and I say so on the chart.)

Case 2 — “Canteen chai sales predict quiz failures”



Rebuild of the published chart. One point per month, $n = 24$. Synthetic data generated for CDB601220 (Week 3).

Here the line fits almost perfectly ($r = 0.98$), so it looks like more chai means more failed quizzes. But both numbers are just counts, and over these two years the number of students tripled.



Both counts divided by students_enrolled. $n = 24$ months; 95% CI for r : -0.12 to 0.62. Synthetic data generated for CDB601220 (Week 3).

When I divide both by how many students were enrolled, most of the link disappears ($r = 0.29$, and with 24 months that could easily be zero). The main thing that pushes failures up per student is exam months, which has nothing to do with chai.

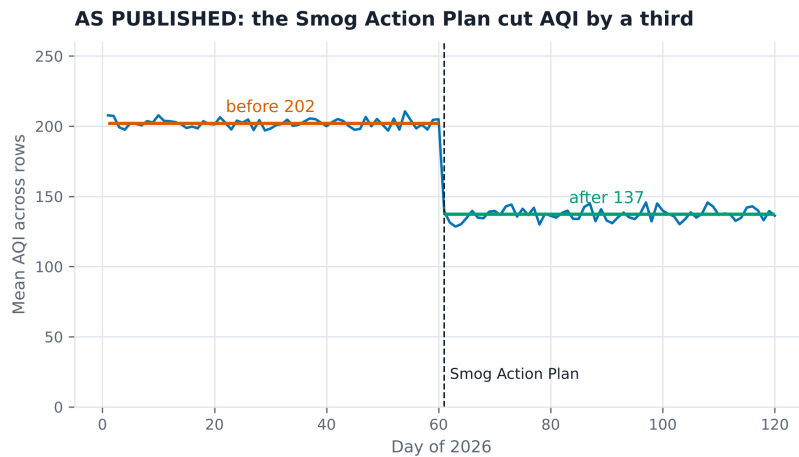
The conclusion it invites. The chart says that months with more doodh patti sold are months with more quiz failures ($r = 0.98$), so chai seems to be bad for marks.

The decoy, and why it does not hold. I first thought exam months: stress makes students drink more chai and also fail more quizzes. But if that were the reason, the link should vanish inside each group. It does not: $r = 0.996$ in non-exam months and $r = 0.996$ in exam months. Exam months are not what is producing the line.

The mechanism. Both counts are driven by head-count. Enrolment grew 3.1x over the 24 months (1,765 to 5,406 students). Cups sold track enrolment at $r = 0.997$ and failures track it at $r = 0.977$. More students means more cups and more failures. When I divide both by students enrolled, the correlation falls from 0.98 to 0.29, and with only 24 months its 95% interval runs from below zero to about 0.6. Per student there is no clear link at all. The one thing that does move failures per student is the exam calendar: 3.86 failures per 100 students in exam months against 3.41 in teaching months, which makes sense, but has nothing to do with chai.

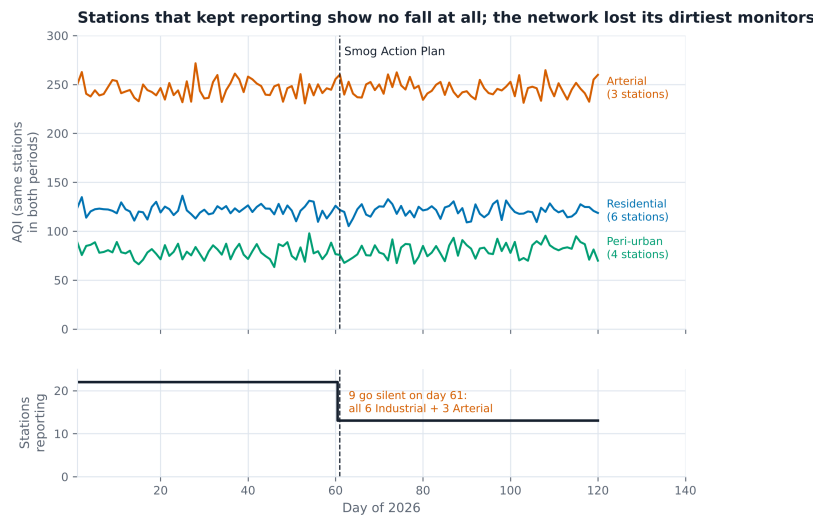
The redesign. A scatter of cups per student against failures per student, with exam months marked by a different shape. The honest title is that per student the two barely move together.

Case 3 — “The Smog Action Plan cut Lahore's AQI by a third”



Rebuild of the published chart, y-axis from zero. Synthetic data generated for CDB601220 (Week 3).

The average AQI falls sharply on day 61, from 202 to 137, so the plan looks like a big success. The axis starts at zero, so this isn't an axis trick. Something else changed on that day.



Top: daily mean AQI by zone for the 13 stations reporting on all 120 days. Industrial has no 'after' data, so it is not drawn. Synthetic data generated for CDB601220 (Week 3).

The bottom panel shows the real reason: on day 61, 9 of the 22 stations stopped reporting, and those were the most polluted ones (all the Industrial stations). The stations that kept reporting stay flat, so air quality didn't actually improve. The average just lost its dirtiest monitors.

The conclusion it invites. The line drops on day 61 from a mean of 202 to 137, so the chart says the Smog Action Plan cut air pollution by 32%.

The decoy, and why it does not hold. The usual objection is a truncated axis. That is ruled out because the recipe made me draw the y-axis from zero, so the fall is not an axis trick. The second objection is the season (winter smog clears by spring). I tested that too: stations that reported in both periods show no fall at all, so there is no seasonal drop in this data either.

The mechanism. The stations that went offline are not random. From day 61, 9 of the 22 stations stopped reporting: all 6 Industrial stations and 3 of the 6 Arterial ones. Before the plan those 9 averaged an AQI of 296, against 137 for the stations that stayed. The daily mean is a mean over whichever stations were reporting, and pandas skips the blanks silently. So the dirtiest monitors dropped out of the average and the average fell. For the 13 stations that reported the whole time, AQI went from 137.1 to 137.2, which is flat (+0.1%). The plan did nothing visible; the network shrank.

The redesign. Two panels. The top one plots AQI per zone using only stations that reported in both periods. The bottom one plots how many stations were reporting each day. The 22 to 13 drop on day 61 is now right under the reader's eyes.

Case 4 — “Four departments behave identically, so fund them equally”

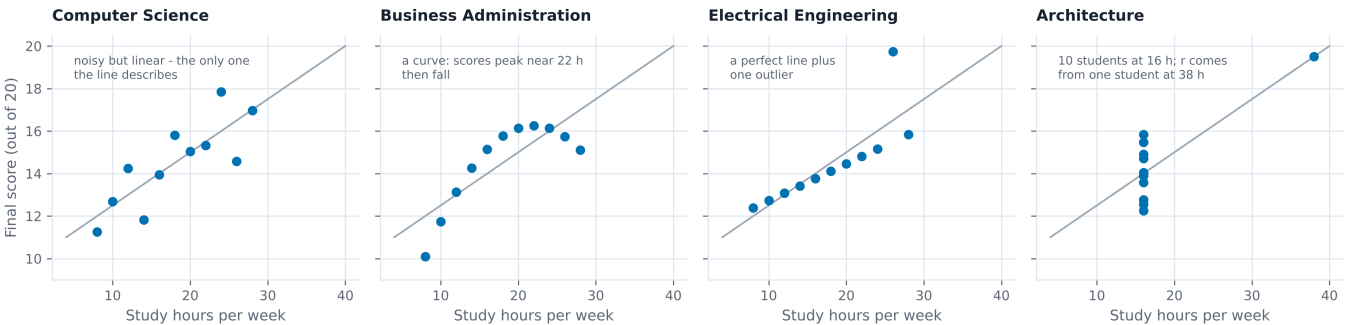
AS PUBLISHED: four departments, identical statistics

	Architecture	Business Administration	Computer Science	Electrical Engineering
n	11	11	11	11
mean study hours per week	18.000	18.000	18.000	18.000
mean final score out of 20	14.501	14.501	14.501	14.500
SD study hours per week	6.325	6.325	6.325	6.325
SD final score out of 20	1.936	1.937	1.937	1.936
Pearson r	0.817	0.816	0.816	0.816
slope	0.250	0.250	0.250	0.250
intercept	10.002	10.001	10.000	10.002

Population SD (ddof = 0). Synthetic data generated for CDB601220 (Week 3).

The table makes the four departments look exactly the same: same means, same SDs, same r and the same slope. Going by these numbers you'd treat all four the same way.

Same mean, SD, r and line - four completely different relationships



Grey line: the shared fit $y = 10 + 0.25x$. 11 students per department. Hours are self-reported. Synthetic data generated for CDB601220 (Week 3).

When I plot the points, the four departments look nothing alike. Only Computer Science really follows a straight line. Business is a curve, Electrical has one outlier, and in Architecture the whole correlation comes from one student. The summary numbers hid all of that.

The conclusion it invites. The table says the four departments are the same: every one has mean 18.00 hours, mean score 14.50, $r = 0.816$ and slope 0.25, so one policy should fit all of them.

The decoy, and why it does not hold. I first thought $n = 11$ per department is too small to say anything. But small samples do not make four groups report identical statistics to two decimals, and once plotted the differences are obvious even with 11 points. The sample size is not the problem; the summary is.

The mechanism. This is Anscombe's quartet. The same numbers come from four completely different shapes. Computer Science is the only department where a straight line is a fair summary. Business Administration is a smooth curve: scores rise to about 22 hours a week and then fall, so more study past that point is linked with lower marks. Electrical Engineering is a perfect line except for one student (26 hours, 19.7). In Architecture, 10 of the 11 students report exactly 16 hours; the whole slope and the whole $r = 0.816$ come from one student who reported 38 hours. Take him out and the correlation cannot even be computed, because x no longer varies.

The redesign. Four small scatter plots on shared axes, with the common regression line in grey. Each department gets its own one-line description. Funding them the same way because their summaries match would be a decision based on four numbers that hide the data.

Case 5 — “Five cohorts, one set of statistics”

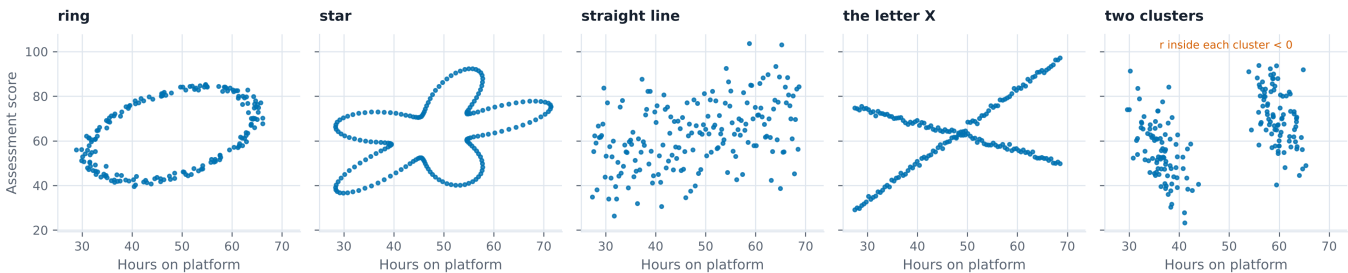
AS PUBLISHED: five cohorts, one set of statistics

	ring	star	straight line	the letter X	two clusters
n	180	180	180	180	180
mean hours on platform	48.000	48.000	48.000	48.000	48.000
mean assessment score	63.000	63.000	63.000	63.000	63.000
SD hours on platform	12.000	12.000	12.000	12.000	12.000
SD assessment score	15.000	15.000	15.000	15.000	15.000
Pearson r	0.420	0.420	0.420	0.420	0.420
slope	0.525	0.525	0.525	0.525	0.525
intercept	37.800	37.800	37.800	37.800	37.800

Population SD (ddof = 0). Synthetic data generated for CDB601220 (Week 3).

Five cohorts with the same mean, the same SD and the same correlation. The numbers were forced to match, so they tell you nothing about what the data actually looks like.

Identical mean, SD and r (0.42) - five shapes that mean five different things



180 students per cohort. Statistics forced with vizlib.match_stats(). Synthetic data generated for CDB601220 (Week 3).

Plotted, the five cohorts are a ring, a star, a line, an X and two clusters, which are completely different patterns. In the two-cluster group the correlation inside each cluster is actually negative. The positive r only comes from the gap between the two groups.

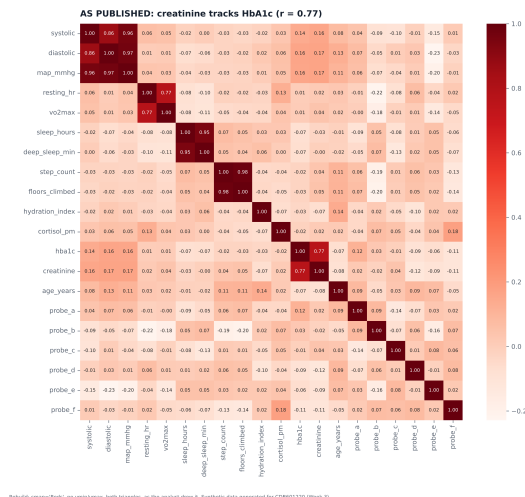
The conclusion it invites. The table says the five cohorts are interchangeable: mean 48.00 hours, mean score 63.00, SD 12.00 and 15.00, $r = 0.420$ in all five.

The decoy, and why it does not hold. With 180 students in each cohort it is tempting to say the numbers are reliable, so the cohorts really are alike. Large n does make these statistics precise, but precise statistics of the wrong summary are still the wrong summary.

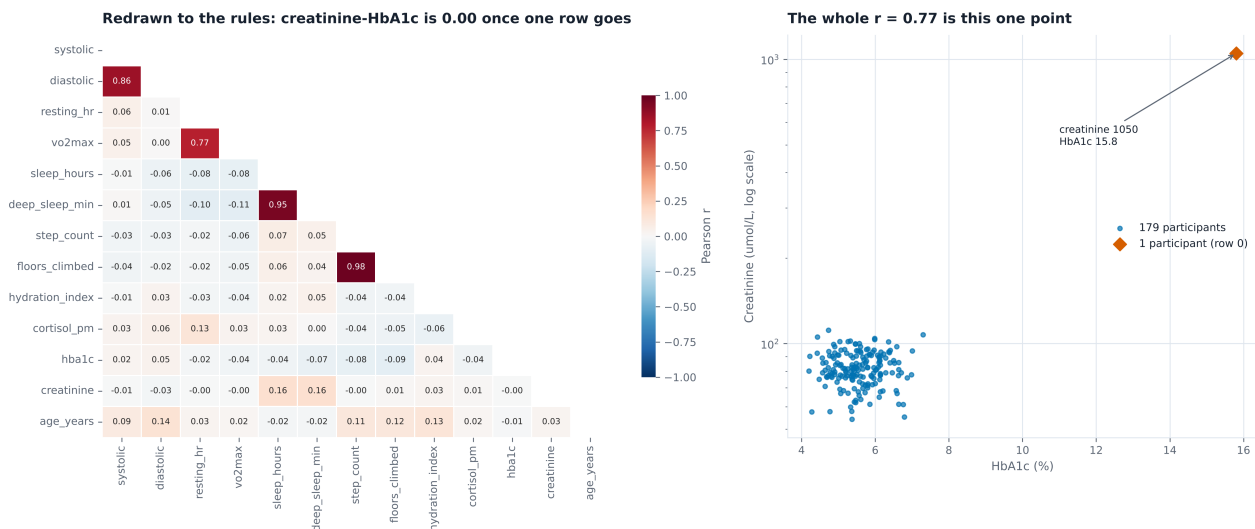
The mechanism. The statistics were forced. vizlib.match_stats() standardises x, removes the part of y explained by x, and rebuilds y with exactly the target r. It keeps the shape, so any shape can be given any mean, SD and r. The clearest proof is the 'two clusters' cohort: inside each cluster the correlation is negative ($r = -0.57$ and -0.45). The whole $+0.42$ comes from the gap between the two clusters, not from more hours going with better scores. The 'ring', 'star' and 'letter X' cohorts have no linear trend a reader would recognise at all, and 'the letter X' is two opposite trends added together. Forcing the statistics also has side effects: in the 'straight line' cohort some scores reach 103.7, which is impossible if the assessment is out of 100.

The redesign. Five small scatter plots on the same axes. The shapes are the result; the table was never enough to describe them.

Case 6 — “Creatinine tracks HbA1c - a finding”



This is the analyst's heatmap. Everything is red, so you can't tell a positive correlation from a negative one, and every number shows up twice. The creatinine–HbA1c value of 0.77 looks like a real finding.



On the left I redrew the heatmap properly with a diverging colour map. On the right you can see the 0.77 comes from one single person with extreme values. Take that one row out and the correlation is basically zero. Fixing the colours alone would not have caught this; I had to look at the actual data.

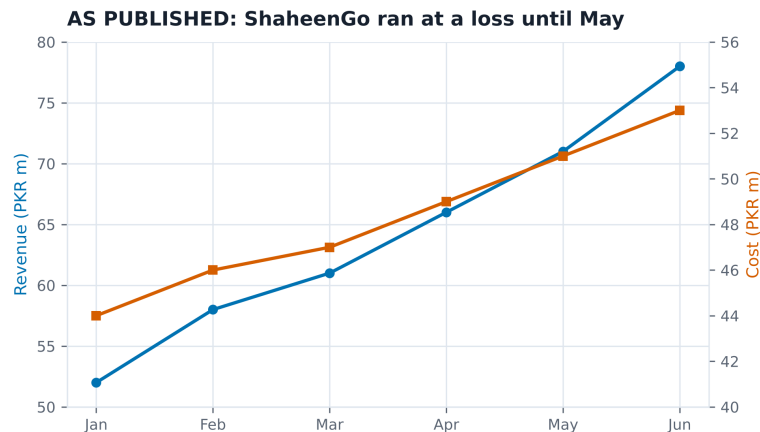
The conclusion it invites. The heatmap says creatinine and HbA1c are strongly linked ($r = 0.77$), which would suggest kidney function and blood sugar move together in this cohort.

The decoy, and why it does not hold. ‘Correlation is not causation’ is true and gets zero, so I tested a real alternative instead: maybe study site drives both. Site C does show $r = 0.92$, but only because it contains one participant. Without that row Site C's r is -0.11.

The mechanism. Two different problems. (1) What it says: the 0.77 is one participant. Row 0 has creatinine 1,050 umol/L (the other 179 are between 54 and 111) and HbA1c 15.8%. Remove that one row and r falls from 0.77 to -0.004; the Spearman correlation, which is not driven by one extreme value, is 0.03. There are other traps in the same matrix: systolic and diastolic vs map_mmhg (0.96, 0.97) are arithmetic because MAP is computed from them; resting_hr vs vo2max is +0.77, the wrong sign for physiology, and inside each heart-rate device model it is negative (-0.40 to -0.63) because the four devices read about 15 bpm apart and were used at different sites; and with 190 pairs, 20 pass $p < 0.05$ when about 9.5 would be by pure chance, and 8 of those ‘hits’ involve the uncalibrated probe channels (probe_e vs diastolic $r = -0.23$). Only 7 pairs survive a Bonferroni correction. (2) How it is drawn: ‘Reds’ is a sequential map for a quantity that has a sign, and without vmin/vmax its lightest colour is the most negative r , so -0.23 and 0 look alike and nothing shows direction. Both triangles show every number twice, and the diagonal of 1s is the darkest thing on the chart.

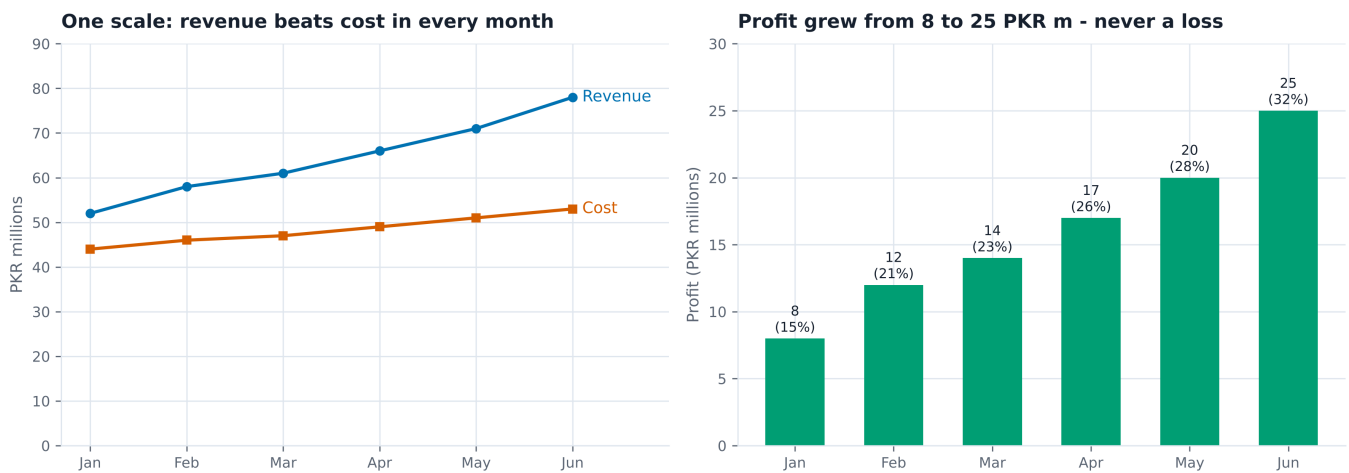
The redesign. I redrew it to the Lecture 6 rules (RdBu_r, -1 to +1, centred at 0, upper triangle masked), dropped map_mmhg and the six probes, and removed the one bad row. I also plot the creatinine vs HbA1c scatter with the outlier marked. Important: fixing the drawing only fixes the sign and redundancy problems. The perfectly drawn heatmap with the outlier still shows 0.77. Only looking at the data fixes what the chart says.

Case 7 — “ShaheenGo ran at a loss until May”



Rebuild: two independent y-axes (twinx), left 50-80, right 40-56. Synthetic data generated for CDB601220 (Week 3).

With two separate y-axes, the cost line sits above revenue until May, so it looks like the company was losing money. But that crossing only happens because of the limits chosen for each axis.



Both series in PKR millions on one axis from zero. Labels: profit (margin). Synthetic data generated for CDB601220 (Week 3).

On one shared axis starting from zero, revenue is above cost every single month. Profit grows from 8 to 25 million PKR, so the company was never running at a loss.

The conclusion it invites. On the dual-axis chart the cost line sits above the revenue line for January to April, so the company looks loss-making until May.

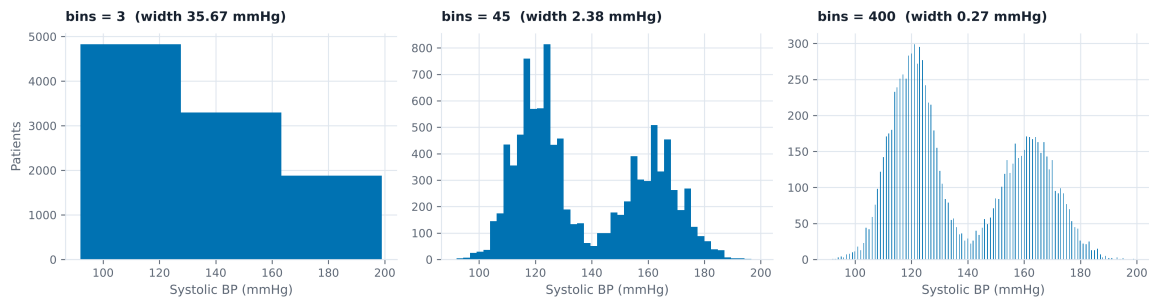
The decoy, and why it does not hold. 'The axes are truncated' is the obvious complaint and it is partly true, but truncation alone cannot make two lines cross. If both series were on one truncated axis, revenue would still sit above cost every month. The truncation is not the mechanism.

The mechanism. Two independent y-axes. Revenue is mapped to 50 to 80 on the left and cost to 40 to 56 on the right, so the vertical position of each line depends only on which limits the analyst picked. Both columns are in the same unit (PKR millions), so there was never a reason for a second axis. On one shared scale revenue is above cost in every month: profit is 8, 12, 14, 17, 20 and 25 PKR m, and the margin grows from 15% to 32%. The company was profitable all along, and more so each month.

The redesign. Revenue and cost on one axis from zero, directly labelled, plus a second panel with monthly profit as bars. Two measures, two panels, no twinx.

Case 8 — “Three bin widths, three different distributions”

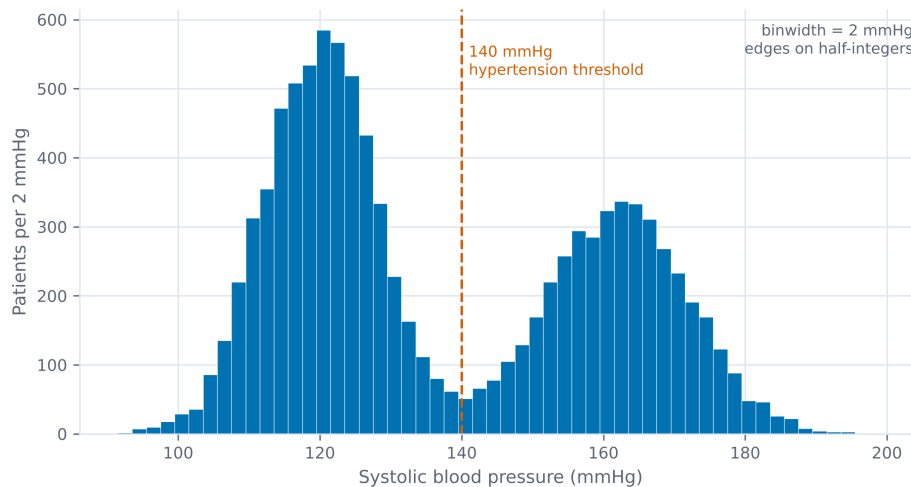
AS PUBLISHED: three bin widths, three different distributions



Rebuild: same 10,000 values, bins = 3 / 45 / 400, no shared axes. Synthetic data generated for CDB601220 (Week 3).

Same 10,000 numbers, three completely different-looking histograms. Only the number of bins changed. 3 bins hide the shape, and 400 bins produce a spiky comb.

Systolic BP is two populations: peaks near 121 and 163 mmHg, 42% above 140



n = 10,000 (no rows removed; systolic has no faults). Synthetic data generated for CDB601220 (Week 3).

With a sensible bin width of 2 mmHg, it's clear there are two groups of patients: one around 121 mmHg and a hypertensive group around 163 mmHg. 42% of patients are above 140. The 400-bin spikes only appeared because blood pressure is recorded in whole numbers.

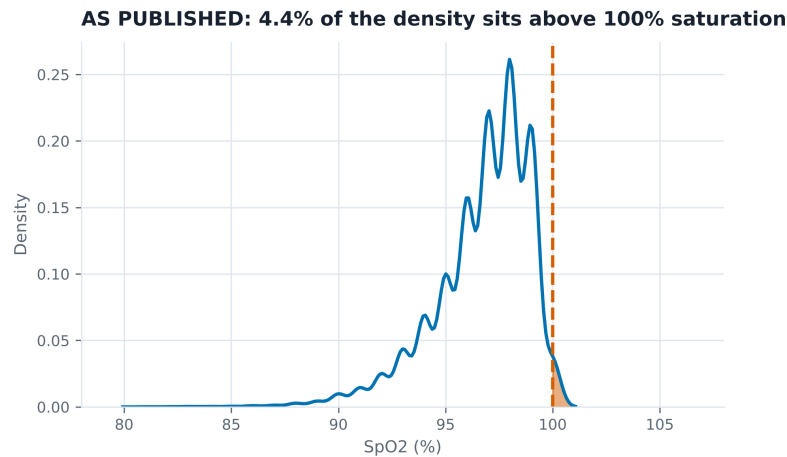
The conclusion it invites. The three panels look like three different distributions: a smooth decline (3 bins), something with two humps (45 bins) and a spiky comb (400 bins). The chart suggests we cannot know the shape.

The decoy, and why it does not hold. I wondered if the spikes were terminal-digit preference (nurses rounding to 0 or 5). They are not: the last digits of systolic are almost perfectly uniform (between 955 and 1,060 readings for each digit 0 to 9). The data is fine; the bins are the problem.

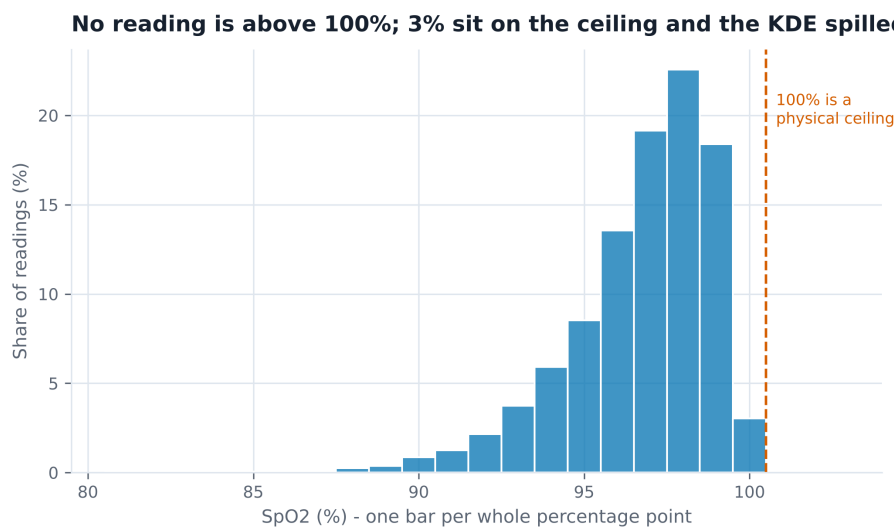
The mechanism. The bin count decides what you see. With bins = 3 each bin is 35.7 mmHg wide, which merges the two populations into one falling staircase (4,829 / 3,294 / 1,877). Systolic is in fact bimodal: one group peaks around 121 mmHg and another around 163 mmHg, with 41.6% of patients above 140. With bins = 400 each bin is only 0.27 mmHg wide, but the readings are whole numbers (104 distinct values), so 296 of the 400 bins are empty. The comb is the integer grid, not noise.

The redesign. One histogram with binwidth = 2 mmHg, edges on half-integers so every bin holds exactly two whole values, and a reference line at 140 mmHg. The bin width is printed on the panel.

Case 9 — “4.4% of the density sits above 100% saturation”



The smooth KDE curve keeps going past 100%, which makes it look like some patients have impossible oxygen levels. But the highest value in the data is exactly 100.



A simple bar for each whole percentage shows the real picture: nothing above 100, and 3% of readings sitting right at the ceiling. The KDE smoothing had spread those values over the limit, so the 'impossible' part of the curve was never in the data.

The conclusion it invites. The curve continues past 100%, so the chart suggests some patients have oxygen saturation above 100%, which is physically impossible and would mean bad data.

The decoy, and why it does not hold. The natural objection is data-entry errors above 100. That is ruled out: the maximum value in the file is exactly 100. No reading is impossible.

The mechanism. The KDE ignores the boundary. spo2 is a whole number with a hard ceiling at 100, and 304 readings (3%) sit right on it. A Gaussian kernel puts a little bell curve on every point, so the bells centred on 99 and 100 spill over the ceiling. With seaborn's default (Scott) bandwidth of 0.35 points, 1.6% of the curve's area is above 100 and 4.3% is above 99.5. The published 4.4% matches the area above 99.5 with the Silverman bandwidth (4.45%), so the analyst counted from the middle of the 100 bin. Whichever way you count it, every bit of that area is created by the smoothing, not by the data.

The redesign. A discrete histogram (one bar per whole percentage point, discrete=True) with the 100% ceiling marked. Nothing can appear above 100 because nothing in the data is above 100. If a smooth curve is wanted, kdeplot(cut=0, clip=(None, 100)) is the minimum fix, and it still has to be declared.